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Structurally innovative vaults in Guarino Guarini's architectural theory: The case of San Gaetano in Vicenza

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ABSTRACT: The seventeenth century architect Guarino Guarini is recognized for his mesmerizing and structurally daring vaults. In addition to his built work, Guarini discussed theoretical matters concerning vaults in the fourth treatise of his *Architettura civile*. Despite being particularly relevant for the design of vaults, the fourth treatise provides hardly anything on actual vault structures. With this in mind, first, this research presents all mentions of actual vault structures in the fourth treatise of *Architettura civile*. Second, Guarini's choice of excluding explicit structural and constructional facets of vaults is discussed in light of his distinction between natural philosophy and experience-based arts. Third, an evaluation of some structural characteristics of Guarini's design for the church of San Gaetano in Vicenza is showcased, arguing that its structural aspects were innovative in Europe at the time, and that they potentially stemmed from similar solutions previously observed in the Middle East and Central Asia.

1 INTRODUCTION

The seventeenth century architect Guarino Guarini (1624-1683) is widely recognized for his mesmerizing and structurally daring vaults. The Chapel of the Holy Shroud in Turin, a project to which Guarini was assigned as "Ducal Engineer" on 19 May 1668 (Meek 1988: 61), is one of the most significant examples in this respect (see Figure 1). In addition to his built body of work, Guarini discussed theoretical matters concerning the general design of vaults – as well as the more detailed issues in this regard – in his book on architecture, *Architettura civile*, published posthumously in 1737. Such aspects of architectural theory are commonly labelled "stereotomy." In the fourth treatise (*Trattato IV*) of the book, Guarini considers these matters by utilizing what one nowadays would deem nascent methods of descriptive geometry to deal with problems of finding true sizes of surfaces of various intersecting geometric solids. In turn, these methods could be useful for, for instance, timber centering, production of templates, and dressing of voussoirs. Guarini had previously presented these methods with a more general scope in his commentary on Euclid's *Elements* (Guarini 1671: 444-452, 572-596). Despite being a segment of a treatise on architecture particularly relevant for the design of vaults, the fourth treatise of *Architettura civile* seemingly has very little space allocated to discussion of actual vault structures. Contrarily, it appears that for Guarini the design of vaults was a purely geometric endeavor. For instance, in "Capo terzo" Guarini discusses "cylindrical surfaces," "oblique cylindrical surfaces," and "plane surfaces" instead of "barrel vaults," "skewed barrel vaults," and "walls" (Guarini 1737: 196-223). Similarly, in "Capo quarto" he deals with "cones" and "conical surfaces" instead of "spires" and "trumpet squinches" (ibid.: 223-244). There are, however, exceptions. For example, in the chapter on "spherical surfaces" (that is, "domes," "Capo quinto," ibid.: 245-254) he briefly explains that some of the proposed geometric procedures are useful for making templates for actual

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Figure 1. Chapel of the Holy Shroud in Turin. Photo: Authors.

stonecutting (ibid.: 246-247). Also of importance (and perhaps equally so with respect to the general design of vaults), albeit implicitly, are the sheets depicting architectural designs by Guarini himself (ibid.: Tavolo 1-34). It has to be noted that the sheets are without accompanying text explaining, for instance, the reasoning behind his choices for particular vaulting structures. That being said, these sheets are arguably the closest one can get to understanding some of the structural aspects related to Guarini's theorizing about vaults.

Since the 1960s, important research has been done on Guarini's stereotomy (related to both theory and practice), ranging from, for instance, Werner Oechslin's and Werner Müller's articles in the proceedings published after the 1968 Guarini conference in Turin (Oechslin 1970; Müller 1970), to several texts in the 2006 Guarini anthology edited by Giuseppe Dardanello, Susan Klaiber, and Henry Armand Millon (Dardanello et al. 2006), and notes on Guarini's theory of spherical vaults by Paolo Borin and José Calvo-López (2020). Even so, structural aspects of vaults in *Architettura civile* has not yet been exhausted by existing scholarship (although specific cases have been treated in part by, for instance, Elwin Robison (1991)).

With this in mind, first, this research will present a systematic lay out of the actual vault structures mentioned in the fourth treatise of *Architettura civile*. Second, Guarini's choice of excluding explicit structural and constructional facets of vaults in *Architettura civile*, both in text and drawings, will be discussed in light of his stated distinction between natural philosophy (that is, similar

to the modern conception of science, for instance, mechanics) and experience-based arts. Third, an evaluation of some of the structural characteristics of Guarini's unbuilt design for the church of San Gaetano in Vicenza (Guarini 1737: Tavolo 26-27) will be showcased. Based on the evaluation, it will be argued that the design has inherent structural aspects that were innovative in a European context at the time, and that they potentially stemmed from similar solutions previously observed in the Middle East and Central Asia.

2 ACTUAL VAULT STRUCTURES IN THE FOURTH TREATISE OF *ARCHITETTURA CIVILE*

Although the fourth treatise of *Architettura civile* has more than seventy pages dedicated to stereotomy, there are altogether only seven brief mentions of actual vault structures therein (ibid.: 191, 203, 246-247, 247, 248, 252, 256). For instance, related to the surface development of a cylinder cut on one side by an oblique plane and on the other by a perpendicular cylinder (ibid.: 201-204), Guarini notes that the procedure may be relevant for template production when constructing a barrel vault (ibid.: 203). Another example is Guarini explaining – with respect to the surface development of a hemisphere posited on a triangular plan (i.e., a triangular sail vault) – that the architect has to be aware of the specific properties of the site, materials, vault shape, and so on when designing (ibid.: 246-247). Some of these examples have been pointed out (but not exhaustively mapped) by Franco Rosso (2006: 310-311). Apart from this, there are no other mentions of actual vault structures in the fourt treatise. It has to be noted that Guarini presented methods for composing idealized geometric vaults elsewhere in *Architettura civile*, but these are not pertaining to structural or constructional issues (Guarini 1737: 183-190).

3 ARCHITECTURAL THEORY VERSUS ARCHITECTURE FOR GUARINI

In the introduction to the first treatise (*Trattato I*) of *Architettura civile*, Guarini wrote that architecture "... is a faculty [*facoltà*], which exercises itself in ordering all sorts of buildings ..." (Guarini 1737: 1, authors's English translation. Unless stated otherwise, all translations are by the authors). He explains that this stance is different from Vitruvius's who describes architecture as a science (*scienza*) or cognition (*cognizione*), and that he is rather in line with what Claude-François Milliet Dechales teaches in Vol. I, Tract. X of his *Cursus seu mundus mathematicus* (ibid.: 1; Dechales 1674). With this in mind, it is crucial to mention that Guarini explained that architecture has "... two functions and is occupied in two ways: one in forming ideas [*idee*], or designing [*disegno*] ... the other is execution [*esequzione*] ..." (Guarini 1737: 2). He specifies that the architect is confined to forming ideas or designing, and relies on the arts – of which he or she is the judge (following Vitruvius, ibid.: 1) – for execution. Considering this, it is not at first glance obvious why Guarini did not follow Vitruvius in defining the theoretical treatment of architecture as a science. From an Aristotelian perspective (Guarini had an education largely founded on the writings of Aristotle, see Klaiber 2006), however, doing architecture would not be an art (*ars*/*techne*, similar to the modern concept of "craft") per se, although it relies on many of them, because then it would pertain to the execution (*esequzione*) of architecture (and the architect would not be an architect but instead, for instance, a carpenter or a stonemason); neither would it be a science (*scientia*/*episteme*), because, as Aristotle pleaded, science is verbally dependent on the strict use of logic (*logos*) (Mitrović 2005: 66-67). Since the work of the architect – that is, according to Guarini, forming ideas, or designing – is an endeavor intrinsically connected to something *visual*, thus depending on geometry (Guarini 1737: 1), it is incompatible with science in the strict Aristotelian conception of it, namely adhering to *words*. Similar to Leon Battista Alberti (Mitrović 2005: 63-68), Guarini attemptively remained true to Aristotelian prerequisites in the search for truth by adjusting the theoretical framework of his system in the sense of denoting architecture as "faculty" instead of "science." The Italian word *facoltà* is a derivative of the Medieval Latin *facultas* (gen. *facultatis*), which means "ability," "capacity," or "skill." By doing so, in *Architettura*

civile, Guarini could theoretically treat the visually dependent discipline of architecture without compromising Aristotelian scientific ideals too extensively.

Although this framework allows Guarini to theoretically treat the practical faculty of architecture, it does not mean that he delves into concrete architectural manifestations, except to a minimal degree, as we have seen. Giorgio Pigafetta and Antonella Mastrorilli's comprehensive study on the relationship between Guarini's mechanical theory in *Placita philosophica* (1665) and his actual constructions in general, conveys an interpretation similar to what shall be employed specifically to the relationship between Guarini's theory of stereotomy and his actual built vaults: "… the theoretical primacy of science over art and experience has the parallel in the primacy of deduction over induction, of the 'principle' over experiment. This is the epistemological knot to clarify before going into the specific treatment of Guarinian science and mechanics to grasp its relationship with the notion of *firmitas*" (Pigafetta, Mastrorilli 1999: 54). Given the few mentions of actual vault structures in Guarini's theory of stereotomy shown above, the interpretation also stands firm with respect to the present topic. Although attempts to make connections between the abstract geometric solids of Guarini's theory of stereotomy and actual vault structures have been made (Calvo-López 2020: 97), there is no evidence in *Architettura civile* supporting that Guarini explicitly made such connections to any considerable degree. Of course, Guarini intended for his stereotomic procedures to be implemented into architectural practice, but he was of the opinion that it is the architect's (and not the architectural theorist's) task to do so (Guarini 1737: 3), as he himself did in an exemplary manner in his built architectural works.

4 STRUCTURAL INNOVATION OF VAULTS IN *ARCHITETTURA CIVILE*: SAN GAETANO IN VICENZA

In the classical tradition of architecture, the interior geometry of domes is very often hemispherical. There are several theological and theoretical reasons for this, but of particular importance is the notion that the shape of the circle, and consequently the sphere when dealing with three-dimensional issues, by several theoreticians have been considered the optimal geometric representation of divinity (Wittkower 1971: 3-13). The most famous example of a hemispherical dome is arguably the Pantheon in Rome (built c. 120 A.D.).

Masonry and ashlar domes carry loads in pure compression (apart from hoop stresses). The geometry of such structures must be so that the compression resultant/thrust line lies within the structure; if the thrust line touches the boundaries of the structure in more than three places, failure occurs (Block et al. 2006). A hemispherical geometry will only be optimal for carrying an evenly distributed radial load. The dominant load in an ashlar dome will, however, be the self-weight of the structure, acting vertically. To compensate for a less-than-optimal shape in the Classical hemispherical dome, one must increase the thickness so that the thrust line stays within the cross section of the structure. The increased thickness means more materials are needed, and thereof increased loads onto the foundations.

The proposed design for the central space of the church of San Gaetano (Guarini 1737: Tavolo 26-27, see Figure 2), dated about 1675 (Klaiber 1994), is an early European example of a discontinuous double-shell dome where the two shells preform distinct and separate statical tasks; the two shells are only connected at the pendentive where they transfer their loads to the underlaying structure, meaning they do not transfer any forces to one another. The inner shell has the geometry of a hemisphere with an oculus, the sole statical task of which is to carry its self-weight down to the supports. The outer shell carries its self-weight and the weight of the spier, in addition to withstanding horizontal forces (such as wind loads). Compared to the inner shell, the shape of the outer shell is closer to a catenary curve, with the sides more straightened out towards the point where the load transfers from the spire – geometrically close to the thrust line. Since the geometrically less optimal inner shell carries relatively small and distributed loads, the increase in thickness to contain the thrust line within the section can be limited. The thickness of the heavier loaded outer shell can also be limited, since the geometry corresponds better with the thrust line.

The discontinuous double-shell dome principle proposed for San Gaetano is a structurally rational solution to the problem of unifying the requirements of having an "architecturally

correct" hemispherical ceiling of the interior space, and at the same time when seen from the outside, have a tall and imposing dome with a heavy spier (alternatively a lantern) on top.

The double-shell principle also has a potential advantage during the construction phase. In the case of San Gaetano, the lighter inner shell could first have been built with the centering going down to the gallery and the supporting arches. With the inner shell established, the outer shell could have been built with the centering supported by the inner shell, thus reducing the loads onto and the height of the centering. In turn, this would have meant that the amount of timber for the centering would have been reduced.

The principle of assigning different statical tasks to separate shells had first been developed in the Middle East and Central Asia during the Medieval Islamic era (970-1600 A.D.) (Ashkan, Ahmad 2010). In Europe, the principle gained traction as a method of building large domes in churches in the years following Guarini's design for San Gaetano. Well-known examples are St. Paul's Cathedral in London (built 1675-1710) and the Pantheon in Paris (built 1758-1790).

As pointed out by, for instance, Meek (1988: 12-13), Guarini's dome designs may have been influenced by Islamic architecture. Although there is still no documentary evidence proving such an influence, his early proposal of using the discontinuous double-shelled dome principle in a European context strengthens the hypothesis.

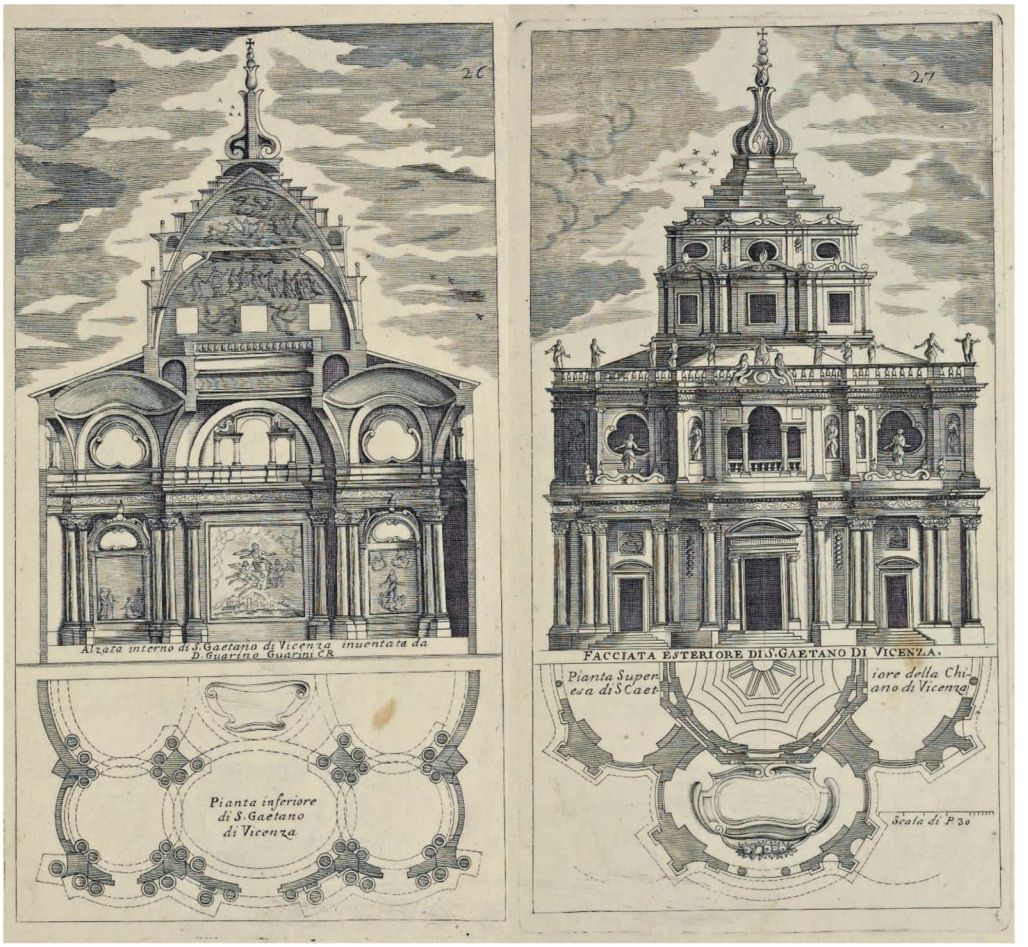


Figure 2. Plans, section, and elevation of the church of San Gaetano, Vicenza (Guarini 1737: Tavolo 26-27). Courtesy Getty Research Institute, Los Angeles (83-B2254).

5 CONCLUSION

This research has shown that the choice by Guarini of including few mentions of explicit structural and constructional facets of vaults in the fourth treatise of *Architettura civile* was rooted in his stated distinction between natural philosophy and experience-based arts. Despite all of this, based on an evaluation of some of the structural characteristics of Guarini's unbuilt design for the church of San Gaetano in Vicenza, it has been argued that the design has inherent structural aspects that were innovative in a European context at the time, and that they potentially stemmed from similar solutions previously observed in the Middle East and Central Asia.

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